



# ApexCeloGrid

An open public ledger for Nigeria's electricity

---

A free, open-source system that records verified electricity generation — from solar mini-grids to any generator in Nigeria — onto a public blockchain that anyone can read, verify, and build on.

**White Paper** · Version 1.0

July 2026

Open-source · Free for everyone · Built for public benefit

# Contents

|                                                       |   |
|-------------------------------------------------------|---|
| 1. Abstract                                           | 2 |
| 2. The Opportunity: Nigeria's Distributed Energy Boom | 2 |
| 3. The Problem: Data Nobody Can Verify                | 3 |
| 4. The Solution: ApexCeloGrid                         | 3 |
| 5. How It Works                                       | 4 |
| 6. The Smart Contract                                 | 4 |
| 7. Why Celo                                           | 5 |
| 8. Free for Everyone                                  | 5 |
| 9. Open Source and Adoption                           | 6 |
| 10. Governance                                        | 6 |
| 11. Roadmap                                           | 6 |
| 12. Impact                                            | 7 |
| 13. Conclusion and Call to Action                     | 7 |
| References                                            | 7 |

## 1. Abstract

ApexCeloGrid is an open-source public ledger that records how much electricity is generated by solar mini-grids and other power systems across Nigeria. Each production record is written to the **Celo public blockchain**, where it is permanent, tamper-evident, and readable by anyone at no cost. The project is deliberately minimal — a small smart contract and an open submission tool — so that it can be built quickly, run cheaply, and adopted by anyone. Its purpose is not profit but public benefit: to give operators, financiers, regulators, and citizens a single set of honest numbers about the electricity Nigeria produces.

## 2. The Opportunity: Nigeria's Distributed Energy Boom

Nigeria is in the middle of a distributed-energy surge. In 2025 the country added roughly **803 MW of solar capacity, about 96% of it off-grid**, and the Rural Electrification Agency (REA) is deploying more than **1,300 solar mini-grids** backed by around \$750 million in public funds, including 250 interconnected mini-grids designed to feed the national grid.

These thousands of small generators produce data every day. But that data is scattered across proprietary dashboards and spreadsheets, none of which an outsider can independently verify. As the sector scales, the absence of a shared, trusted record becomes a growing bottleneck.

| Indicator (2024–2026)   | Figure                                                     | Why it matters here                                          |
|-------------------------|------------------------------------------------------------|--------------------------------------------------------------|
| Solar added in 2025     | ~803 MW, ~96% off-grid                                     | A large, growing base of generators to record                |
| REA mini-grid programme | 1,300+ mini-grids; ~\$750m; 250 interconnected             | A natural first cohort for a public ledger                   |
| Regulatory framework    | NERC Mini-Grid Regulations 2023 (limits raised to 5–10 MW) | Formal basis; larger assets raise the value of verified data |
| Access reach            | DARES programme: 4.1m+ people reached                      | Strong public-interest case for transparency                 |

Figures from public 2024–2026 reporting; see References. Refresh at project kick-off.

### 3. The Problem: Data Nobody Can Verify

The problem is not measurement — operators already meter their output. The problem is **trust and portability**. A production figure lives inside one operator's system. A bank verifying a results-based payment, a regulator confirming a plant is active, a carbon buyer, or a citizen all have to take that number on faith or pay for a bespoke audit.

This friction slows disbursements, makes clean-energy certificates costly and easy to double-count, and leaves the country without a clear, queryable picture of its own distributed generation. What is missing is a neutral record that no single party owns and everyone can check.

### 4. The Solution: ApexCeloGrid

ApexCeloGrid records verified production onto a **public blockchain that already exists**, so the record is open, permanent, and free to read. It is built on three commitments:

**Free for everyone.** Free to read and free to take part. No operator, agency, or citizen is ever charged to use it.

**Simple to adopt.** It works with the metering equipment operators already have. Joining requires no special hardware and no blockchain expertise.

**Fully open source.** The code is public under a permissive licence. Anyone may use it, copy it, or build on it.

Crucially, ApexCeloGrid is **not a new blockchain**. Building a blockchain is a large, multi-team undertaking. Deploying a small contract onto an established public chain is a job a single developer can ship — and it inherits the openness and durability of a network already run by thousands of computers worldwide.

## 5. How It Works

The system is intentionally small. Data flows from a meter, through a simple tool, into one public contract — and out to anyone who wants to read it.

```
Generation system / meter
| a signed reading: site, period, kWh
v
Submission tool (open-source, free to run)
| one small transaction
v
Celo public blockchain -- ProductionLedger contract (append-only)
|
+--> Anyone reads it: block explorer · public API · ApexCeloGrid website
```

### 5.1 What goes on-chain (and what does not)

Only compact, high-value summaries are stored on-chain — for example, the energy a site produced in a day. Raw high-frequency meter telemetry stays off-chain and is referenced by a cryptographic fingerprint, keeping the ledger small, cheap, and privacy-respecting. No personal or household-consumption data is ever placed on-chain.

| Field                   | Meaning                                                       |
|-------------------------|---------------------------------------------------------------|
| siteId                  | Which generation site produced the power                      |
| periodStart / periodEnd | The time window covered by the record                         |
| energyWh                | Energy produced in the window (whole watt-hours, no rounding) |

## 6. The Smart Contract

The core is one small, append-only Solidity contract. This is the entire first version:

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;

contract ProductionLedger {
```

```

struct Record {
    bytes32 siteId;
    uint64  periodStart;
    uint64  periodEnd;
    uint256 energyWh;
}
Record[] public records;

event Recorded(bytes32 indexed siteId, uint64 periodStart,
               uint64 periodEnd, uint256 energyWh, uint256 index);

function submit(bytes32 siteId, uint64 periodStart,
                uint64 periodEnd, uint256 energyWh) external {
    records.push(Record(siteId, periodStart, periodEnd, energyWh));
    emit Recorded(siteId, periodStart, periodEnd, energyWh, records.length - 1);
}

function total() external view returns (uint256) { return records.length; }
}

```

Everything else — a registry of sites, signed readings for stronger trust, access rules, or a renewable-energy-certificate module — is optional and can be added later, only if the project is adopted. The guiding principle is: **ship the smallest useful version first.**

## 7. Why Celo

---

The requirements — public, free to read, extremely low cost, and credible on sustainability — point to a public, low-fee, energy-efficient (proof-of-stake) EVM chain. Celo fits precisely:

- **Sub-cent fees.** Recording a record costs a tiny fraction of a naira.
- **Carbon-negative.** Celo offsets more carbon than it emits — fitting for a clean-energy registry.
- **Mobile-first.** Designed for the kind of low-bandwidth, phone-based access common in Nigeria.
- **Ethereum-secured.** Since 2025, Celo runs as an Ethereum Layer 2, inheriting Ethereum's security.
- **EVM / Solidity.** Standard tooling, and portable — the same contract can run on Polygon or another EVM chain if ever needed.

## 8. Free for Everyone

---

"Free for everyone" is the non-negotiable requirement, and the design delivers it at every level:

- **Readers pay nothing, ever.** Public explorers, a free API, and the ApexCeloGrid website expose all data openly.
- **Operators pay nothing to take part.** The tiny transaction fee can be sponsored by a shared treasury (a "paymaster"), so an operator never holds a token, touches an exchange, or sees a bill.

- **No coin, no speculation.** ApexCeloGrid has no token of its own. It is a public utility, not a financial product.

**An honest note on cost.** "Free for everyone" means no user is ever charged — not that the shared backbone is costless. Servers and maintenance are funded by grants, public support, or optional premium services for commercial users (such as certificate issuance), so that ordinary access stays free forever. This mirrors how public goods like Wikipedia and OpenStreetMap remain free to all while a foundation funds the infrastructure.

## 9. Open Source and Adoption

All code is published on a public repository under a permissive licence (MIT or Apache-2.0). A suggested structure:

```
apexcelogrid/  
  contracts/    ProductionLedger.sol (+ tests)  
  submit/      open-source submission tool  
  web/         public site: read records + docs + repo link  
  docs/        this white paper + technical spec  
  README.md    what it is, how to run it, how to adopt it  
  LICENSE
```

The author's role ends at "published and documented." From there, **anyone can adopt or fork it** — an operator, a state agency, a university, or a developer — and run it for their community. The clearer and simpler the project, the more likely that adoption happens.

## 10. Governance

ApexCeloGrid stores public data on a public chain, so it needs very little governance. Read access is open and permissionless by design. Where stronger trust is wanted later — for example, confirming that a submitting site is genuinely accredited — a lightweight, multi-stakeholder approach is recommended: a small group that may include the regulator or REA, mini-grid operator associations, and independent maintainers, responsible only for accreditation standards. **No party can edit or delete production records**; the ledger is append-only, and corrections are made by adding a linked adjustment, never by rewriting history.

## 11. Roadmap

| Phase       | Focus                                      | Outcome                            |
|-------------|--------------------------------------------|------------------------------------|
| 1 · Build   | Contract + submission tool on Celo testnet | Working prototype, free to test    |
| 2 · Publish | Public repo, website, white paper, docs    | Anyone can find, read, and fork it |

|              |                                                                    |                                              |
|--------------|--------------------------------------------------------------------|----------------------------------------------|
| 3 · Launch   | Deploy to Celo mainnet; onboard first volunteer sites              | Real production records, publicly verifiable |
| 4 · Adoption | Operators/agencies run it; optional registry & certificate modules | A growing national record — driven by others |

## 12. Impact

---

- A single, trusted public record of distributed generation in Nigeria.
- Faster, cheaper verification for results-based finance and clean-energy certificates.
- Transparency for citizens about the electricity produced in their communities.
- A reusable open-source foundation others can extend across the region.

## 13. Conclusion and Call to Action

---

ApexCeloGrid is a small idea with a large purpose: give everyone the same honest numbers about the electricity Nigeria produces, for free, forever. It is intentionally simple to build, cheap to run, and open to all.

**This is an open contribution to the country. Whoever adopts, forks, or builds on ApexCeloGrid is asked to do one thing in return: use it to help the people of Nigeria.**

## References

---

1. NERC — Mini-Grid Regulations, 2023. [nerc.gov.ng](https://nerc.gov.ng)
2. Ecofin Agency — Nigeria adds 803 MW of solar in 2025, off-grid dominant. [ecofinagency.com](https://ecofinagency.com)
3. Industrial Info — Nigeria's \$750m mini-solar-grid strategy. [industrialinfo.com](https://industrialinfo.com)
4. Mondaq — Understanding I-RECs and Nigeria's clean-energy transition. [mondaq.com](https://mondaq.com)
5. Celo Documentation — Why Celo (Ethereum L2, carbon-negative, low fees). [docs.celo.org](https://docs.celo.org)

This white paper is an informational document about an open-source project. It is not investment, legal, or financial advice.